

TARGET GATE 2026

SUPER 20

SERIES

Database Management System

CS & DA

YOUR TOP RANK SHORTCUT



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#Q. Consider the following instance of CelestialReport (CelestialObject, ReportBy, ReportScore, GroupLead).
key.
[candidate key other than primary key]

CelestialObject	ReportBy	ReportScore	GroupLead
Pluto	Team 5	90	L. Kurtz
Jupiter	Team 5	88	L. Kurtz
Sirius A	Team 1	96	Samuel
Rigel	Team 8	96	Louie

Ans: A, D.

Which of the following is a Secondary Key in CelestialReport?

- A** {ReportBy, ReportScore}. ✓
- B** {ReportBy, GroupLead}. ✗
- C** {GroupLead, CelestialObject}. ✗
key.
- D** {ReportScore, GroupLead}. ✓

#Q. Which of the following can be a candidate key for the following instance?

BookDetails					Ans: (C, D)
BOOK_ID	YEAR_PUB	BOOK_TITLE	AUTHOR_NAME	TOTAL_SOLD_COPIES	
1001 ✓	2010	DBMS	KORTH	15000	
1002 ✓	2010	OPERATING SYSTEM	GALVIN	25000	
1003 ✓	2020	COMPILER	ULLMAN	12000	
1004 ✓	1995	DATA STRUCTURES	CORMEN	20000	
1005 ✓	1995	DATA STRUCTURES	ULLMAN	20000	
1006 ✓	1990	COMPUTER NETWORKS	FOROUZAN	15000	
1007 ✓	2010	MACHINE LEARNING	MITCHELL	18000	

A {BOOK_ID, YEAR_PUB} ✗

B {BOOK_ID, BOOK_TITLE} ✗

C {BOOK_TITLE, AUTHOR_NAME} ✓

D {BOOK_ID} ✓

key

A

$A \cup \emptyset$

$= A$

candidate key

$A \cup B$

$= A|B \checkmark$

$A \cup C$

$= A|C$

$A \cup B C$

$= A|B,C \checkmark$

#Q. Consider the following instance of the relation BookDetails.

BookDetails (BOOK_ID, YEAR_PUB, BOOK_TITLE, AUTHOR_NAME, TOTAL_SOLD_COPIES)

BookDetails				
BOOK_ID	YEAR_PUB	BOOK_TITLE	AUTHOR_NAME	TOTAL_SOLD_COPIES
1001	2010	DBMS	KORTH	15000
1002	2010	OPERATING SYSTEM	GALVIN	25000
1003	2020	COMPILER	ULLMAN	12000
1004	1995	DATA STRUCTURES	CORMEN	20000
1005	1995	DATA STRUCTURES	ULLMAN	20000
1006	1990	COMPUTER NETWORKS	FOROUZAN	15000
1007	2010	MACHINE LEARNING	MITCHELL	18000

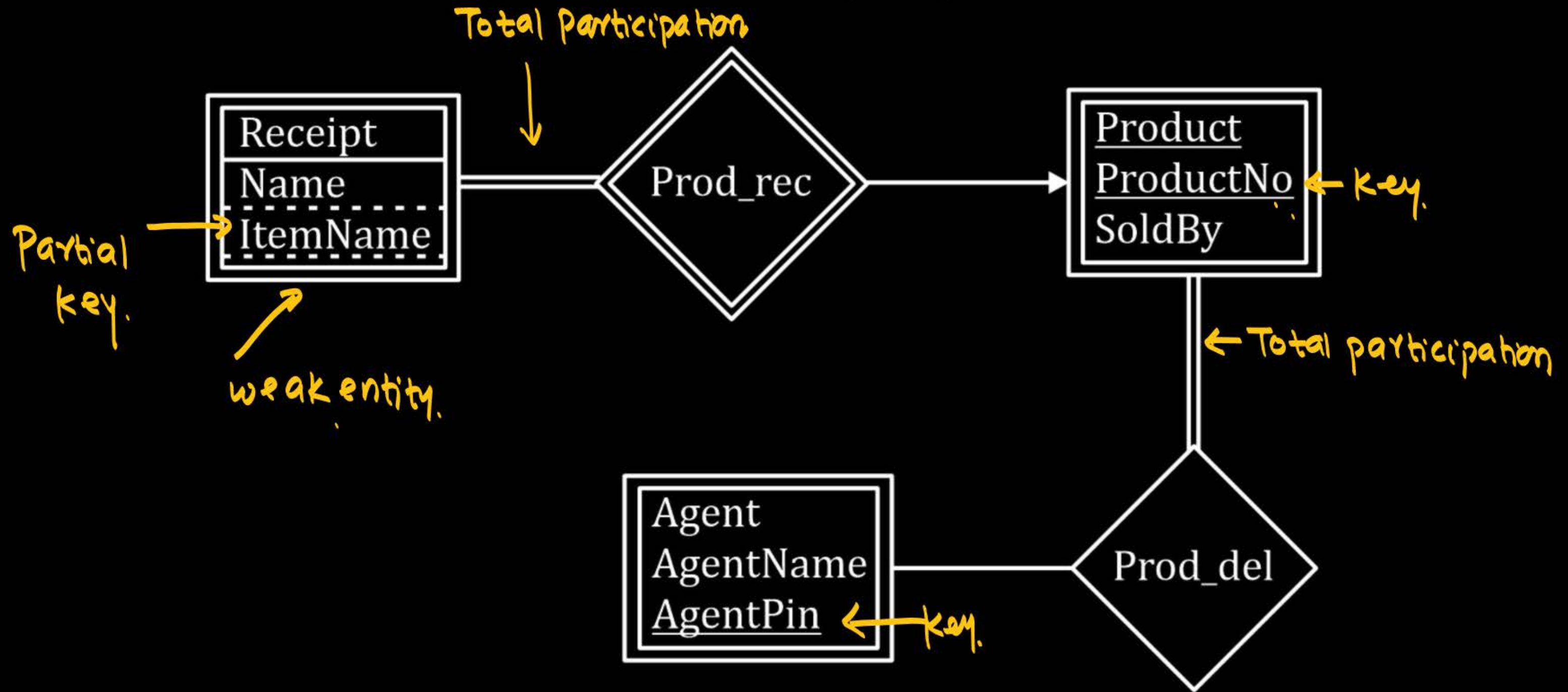
Identify the correct statement(s) to get the following output:

BookDetails				
BOOK_ID	YEAR PUB	BOOK_TITLE	AUTHOR_NAME	TOTAL_SOLD_COPIES
1001	2010	DBMS	KORTH	15000
1002	2010	OPERATING SYSTEM	GALVIN	25000
1003	2020	COMPILER	ULLMAN ,	12000
1005	1995	DATA STRUCTURES	ULLMAN '	20000
1007	2010	MACHINE LEARNING	MITCHELL	18000

- A** SELECT * FROM BookDetails
WHERE YEAR_PUB=2010 AND AUTHOR_NAME='ULLMAN';
- B** SELECT * FROM BookDetails
WHERE YEAR_PUB=2010 OR AUTHOR_NAME='ULLMAN'; ✓
- C** (SELECT * FROM BookDetails
WHERE YEAR_PUB=2010)
INTERSECT
(SELECT * FROM BookDetails WHERE AUTHOR_NAME='ULLMAN');
- D** (SELECT * FROM BookDetails
WHERE YEAR_PUB=2010)
UNION
(SELECT * FROM BookDetails WHERE AUTHOR_NAME='ULLMAN'); ✓

[Ans: B, D]

#Q. Consider the following Entity Relationship Diagram:



Which of the following options is (are) true?

Key for weak entity.

= owner primary key + partial key.

- A** Participation of Product is total in **Prod_del**(τ)
- B** The primary key in the relational schema for Receipt is {**Name, ItemName, (** τ **) ProductNo**}.
- C** The primary key in the relational schema for Receipt will be (**Name, ItemName, AgentPin**).
- D** Participation of Product is total in **Prod_rec**

[Ans: A, B]

#Q. Consider the instance of the relational schema Gallery (PaintingID, Artist, Theme):

Gallery		
PaintingID	Artist	Theme
A12	J.Ray	Landscape
A187	J.Ray	Portrait
B3	KatieP	Abstract
H23	L.Houston	Landscape

A12, J.Ray Portrait

Which of the following Relational Algebra expressions produce(s) exactly the same tuples as present in this instance of Gallery?

- A** $\Pi_{\text{PaintingID, Artist}}(\text{Gallery}) \bowtie \Pi_{\text{PaintingID, Theme}}(\text{Gallery})$
- B** $\Pi_{\text{PaintingID, Artist}}(\text{Gallery}) \bowtie \Pi_{\text{Artist, Theme}}(\text{Gallery})$
- C** $\Pi_{\text{PaintingID, Theme}}(\text{Gallery}) \bowtie \Pi_{\text{PaintingID, Artist}}(\text{Gallery})$
- D** $\Pi_{\text{PaintingID, Theme, Artist}}(\text{Gallery}) \bowtie \Pi_{\text{Theme}}(\text{Gallery})$

Generate extra tuple.

X.

X.

#Q. In a Drawing Competition, **participants** individually can enroll their names for the competition. There are many **groups** in the competition depending on the age of the participants. A **participant** can **participate** in only one **group** and a group consists of multiple **participants**. Each group has a unique name (GName) and a participant also has a unique id. The result (score) of each group is maintained separately. Which of the following statement(s) are **incorrect**?

Ans: A, C.

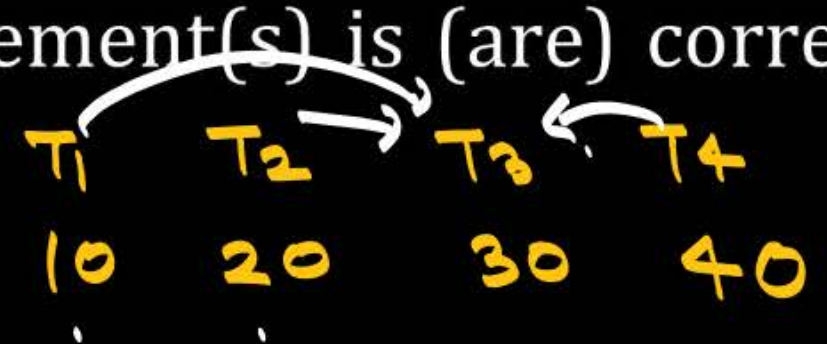
- A** Entity Group does not have any primary key. (F) ✓
- B** Participate is a one-to-many relationship between Group and Participants. ✓ (T)
- C** Participate is a many-to-many relationship between Group and Participant. (F) ✓
- D** GName may be the foreign key of Participate relation between Group and Participant. (T)

#Q. Choose the **incorrect option(s)** about indexing from the following.

- A** There can be many clustering indices in a database file. (F)
- B** There can be at most one secondary index in a database file. (F)
- C** It is preferable to use a dense index when the file is not sorted on the indexed field.
- D** Dense index is preferred over the sparse index when the size of a file is small.

Ans: A, B.

#Q. Suppose in a database, there are four transactions T_1, T_2, T_3 and T_4 with timestamp 10, 20, 30 and 40 respectively. T_3 is holding some data items which T_1, T_2 and T_4 are requesting to acquire. Which of the following statement(s) is (are) correct in respect of Wait-Die Deadlock Prevention scheme?



A Transaction T_1 and T_4 will wait for T_3 to release the data item.

Older $\rightarrow$ wait

B Transaction T_1 and T_2 will wait for T_3 to release the data item. ✓

C Transaction T_2 and T_4 will wait for T_3 to release the data item.

Younger $\rightarrow$ Rollback.

D Transaction T_4 will rollback. ✓

[Ans: B, D]

#Q. Consider the following two schedules S1 and S2.

S1	
T ₁	T ₂
R(X)	
W(X)	
	R(X)
	W(X)
	COMMIT
R(Y)	

S2	
T ₁	T ₂
R(X)	
W(X)	
COMMIT	
	R(X)
	W(X)
	COMMIT

T₂ → Read before T₁ commit.

T₂ → commit T₁.

S₁ is not

R(X) denotes read operation on data item X by Transaction T_i.

W(X) denotes write operation on data item X by Transaction T_i.

Which of the following statement(s) is/are true for the above two schedules S1 and S2?

[Ans: C, D]

- A** Both schedules S1 and S2 are Recoverable Schedule.
- B** Both schedules S1 and S2 are Cascadeless Schedule.
- C** The schedule S1 is not a Recoverable Schedule but the schedule S2 is Cascadeless ✓
Schedule.
- D** The schedule S1 is not a Recoverable Schedule but the schedule S2 is Recoverable ✓
Schedule.

#Q. Assume an immediate database modification scheme. Consider the following log records for transactions T0, T1, T2, T3 and T4:

Steps	Details of log
1	<T0, start>
2	<T1, start>
3	<T1, X, 100, 200>
4	<T1, commit>
5	<checkpoint {T0}>
6	<T0, Y, 200, 400>
7	<T2, start>
8	<T0, commit>
9	<T3, start>
10	<T4, start>
11	<T3, Z, 400, 600>
12	<T4, W, 600, 1200>
13	<T4, commit>

T0, Y, 200, 400

Redo Y = 400

T3, Z, 400, 600

Undo Z = 400

Step 4 → T1 commit

Step 8 → T0 commit

Step 13 → T4 commit

T1, X, 100, 200

Redo X = 200

T4, W, 600, 1200

Redo W = 1200

If there is a crash just after step 13 and the recovery of the system is successfully completed, identify the correct actions for the above scenario.

- A** After recovery completion, value of X will be 100. (F)
- B** After recovery completion, value of Y will be 400. (T)
- C** After recovery completion, value of Z will be 600. (F)
- D** After recovery completion, value of W will be 1200. (T)

[Ans: B, D]

#Q. Consider the following relational schema:

Patients (pid, pname, age, p_email)

Doctors (did, dname, qualification, d_email)

Appointments (pid, did, date, time)

Four relational algebra queries are given below:

Q1: $(\sigma_{\text{age} > 20}(\sigma_{\text{date} = '2022-06-12'}(\text{Patients} \bowtie \text{Appointments})))$

Q2: $\sigma_{\text{date} = '2022-06-12'}(\sigma_{\text{age} > 20}(\text{Patients} \bowtie \text{Appointments}))$

Q3: $\Pi_{\text{dname}}(\Pi_{\text{dname}, \text{d_email}}(\text{Doctors} \bowtie \text{Appointments}))$

Q4: $\Pi_{\text{dname}, \text{d_email}}(\text{Doctors} \bowtie \text{Appointments})$

Identify the correct options from the options given below.

A Q1 is equivalent to Q2. ✓

[Ans: A, D]

B Q1 is not equivalent to Q2

C Q3 is equivalent to Q4

D Q3 is not equivalent to Q4 ✓



Thank
You

